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Covert Channel Based On IP Packet Order Phases 3 & 4: DETECTOR & MITIGATOR

CENG519 Network Security
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1 Introduction

This is the report for the third and fourth phases of the term project for Network Security. A detector and a mitigator for the covert channel based on IP packet order were implemented.

2 Background

2.1 Reorder Density

RFC 5236 [1] defines several metrics for evaluating packet reordering, including Reorder Density (RD). RD measures the extent of packet reordering by quantifying how far reordered packets have moved from their original sequence. Specifically, it sums the differences between an out-of-order packet's sequence number and the sequence number of the packet that immediately preceded it in the received sequence. Higher RD values indicate more significant reordering. This metric is particularly useful for understanding the magnitude of disorder in a packet stream.

3 Detection

The detector was designed to detect the covert channel by measuring the reorder density of the received packets. The detector is implemented in the go-processor, which processes the packets as they arrive. The implementation uses a buffer to store the sequence numbers of received packets, and calculates the reorder density map in real-time, then logs the results of the detection for each frame of packets.

The detection is based on the following heuristics:

- If the reorder density of the first displacement (-1 or +1) is above 0.15, or
- If the reorder density of the zeroth displacement (i.e. packets that arrived in order) is below the 0.60,

then the packets are considered to be reordered, and the covert channel is detected.

These heuristics were chosen based on the literature reviewed in [2], which suggests that the observed reorder density of first differences is usually below 0.15, and that higher rates of packets arrive in their expected order.

4 Mitigation

The mitigator is designed to effectively prevent covert communication for the IP order based covert channel. While aiming to reduce the channel capacity would be an accepted goal, another approach is used here, which is reordering the packets according to their original order by their sequence numbers. This effectively removes the covert channel, although the resulting channel capacity might not be zero, as the original order may still be used as a valid permutation by the covert channel.

The implementation was made by adding a buffer window in the go-processor, which stores the packets until they are ready to be sent in the correct order. If the window is full, the packets are sent in their sorted order, and the window is cleared. The window size is set to 10 packets to avoid excessive delays, and the packets are sorted by their sequence numbers.

5 Results and Discussion

5.1 Methodology

5.1.1 Detection

The detector was tested by sending packets with a fixed delay of 20ms and a jitter of 5ms. The parameters k and bpS were varied, and the reorder density was measured for different values of these parameters. Additional tests were conducted without the covert channel active in order to evaluate the false positive rate of the detector. The tests were conducted with a total of 5000 packets sent at intervals of 10ms, and each frame consisted of 50 packets.

5.1.2 Mitigation

The mitigator was tested under the same conditions as the capacity tests in Phase 2, with a fixed delay of 20ms and a jitter of 5ms. The parameters k and bpS were varied, and the channel capacity was measured for different values of these parameters. The mitigator was active in the go-processor, providing live mitigation.

5.2 Results

5.2.1 Detection

The detection results are shown in Table 1. The table shows the true positive (TP), false positive (FP), true negative (TN), and false negative (FN) rates for different values of k and bpS . The F1-Score is also calculated for each case, which is a measure of the detector's accuracy. Note that the results are for testing whether a covert channel exists or not, and each row represents the tested configuration.

Since the tests are organized based on covert channel parameters, only the first row is expected to have any true negatives, and the others should not have any false positives (as the covert channel is always active).

The difference in detection rates across the configurations shows the potential for stealthiness of the covert channel. Configurations with lower bits per symbol (i.e. lower effective channel capacity) are less likely to be detected, as they result in less significant displacements. For example, the least detected configuration is $k = 4$ and $bpS = 1$, which has the lowest channel capacity of 0.25 bits per packet, compared to the most detected configuration, $k = 4$ and $bpS = 4$, which has a channel capacity of 1 bit per packet.

Parameters	TP	FP	TN	FN	F1-Score
No covert channel	0	0	1.0	0	N/A
$k=2, bpS=1$	0.77	0	0	0.23	0.8701
$k=3, bpS=1$	0.88	0	0	0.12	0.9312
$k=3, bpS=2$	0.86	0	0	0.14	0.9198
$k=4, bpS=1$	0.66	0	0	0.34	0.7952
$k=4, bpS=2$	0.85	0	0	0.15	0.9198
$k=4, bpS=3$	0.90	0	0	0.10	0.9474
$k=4, bpS=4$	0.91	0	0	0.09	0.9529

Table 1. Detection Results

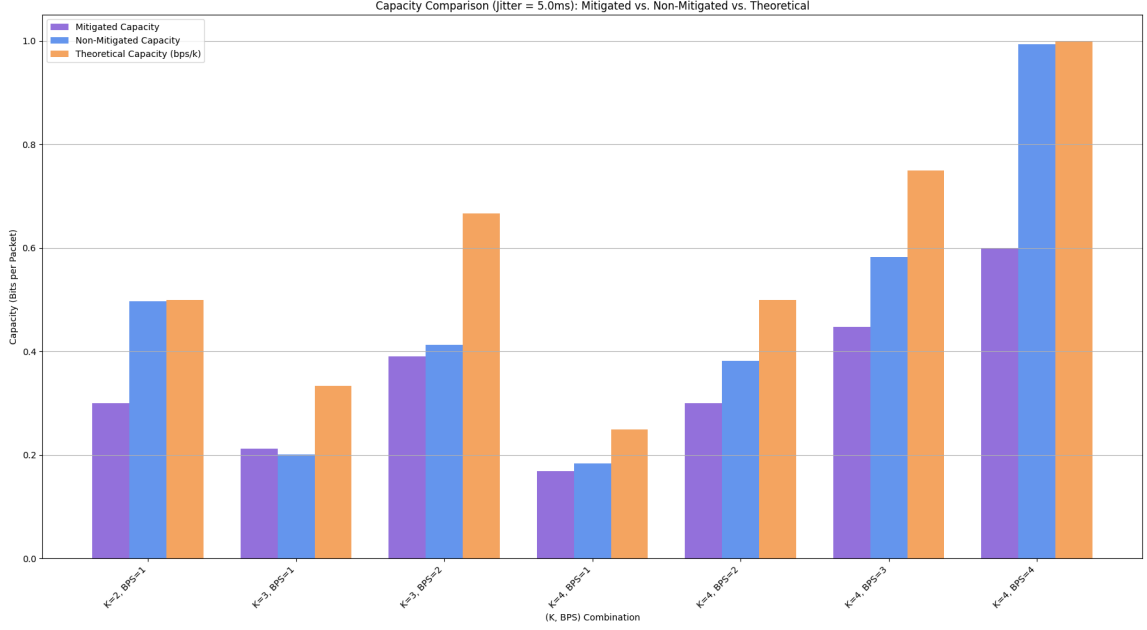


Figure 1 . Mitigated channel capacity for different k and bpS values

5.2.2 Mitigation

Figure 1 shows the mitigated channel capacity for different k and bpS values, along with the non-mitigated and theoretical capacities for comparison, and the bit error rate is shown in Figure 2 .

It can be seen that the mitigator is effective in reducing the channel capacity in general, and is more effective in increasing the bit error rate. For the tested scenarios, the resulting received packet order was always the original order, which prevents the receiver from correctly interpreting the packets as a covert channel.

6 Conclusion

The covert channel based on IP packet order was successfully detected and mitigated. The packet reorder metrics were effective in identifying the covert channel activity. The results demonstrated the effectiveness of the channel in evading detection and the ability of the mitigator to obstruct the covert communication.

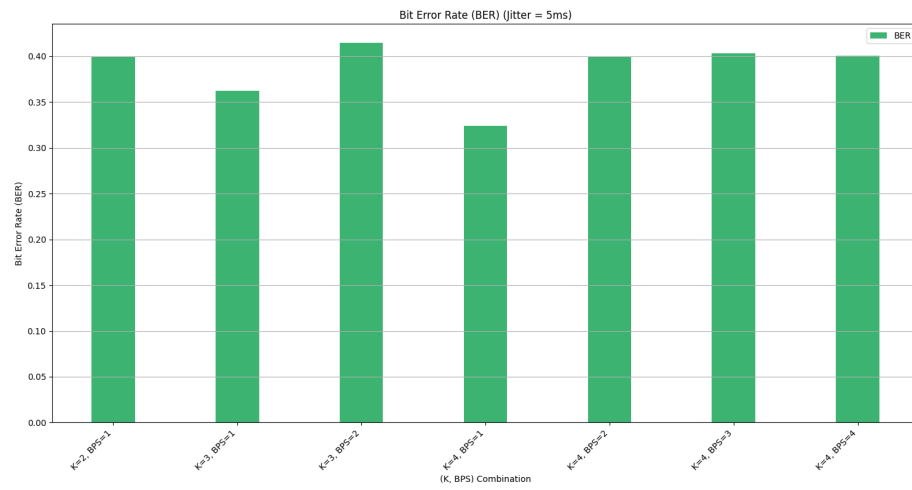


Figure 2 . Mitigated bit error rate for different k and bpS values

References

- [1] *RFC 5236: Improved Packet Reordering Metrics*, Jun. 2008. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc5236>.
- [2] A. El-Atawy and E. Al-Shaer, “Building Covert Channels over the Packet Reordering Phenomenon,” in *IEEE INFOCOM 2009*, Rio de Janeiro, Brazil: IEEE, Apr. 2009, pp. 2186–2194, ISBN: 978-1-4244-3512-8. DOI: 10.1109/INFCOM.2009.5062143.